Lab 7

Fuzzing for Input Validation Bugs

L6_L7_42 - Salon Booking

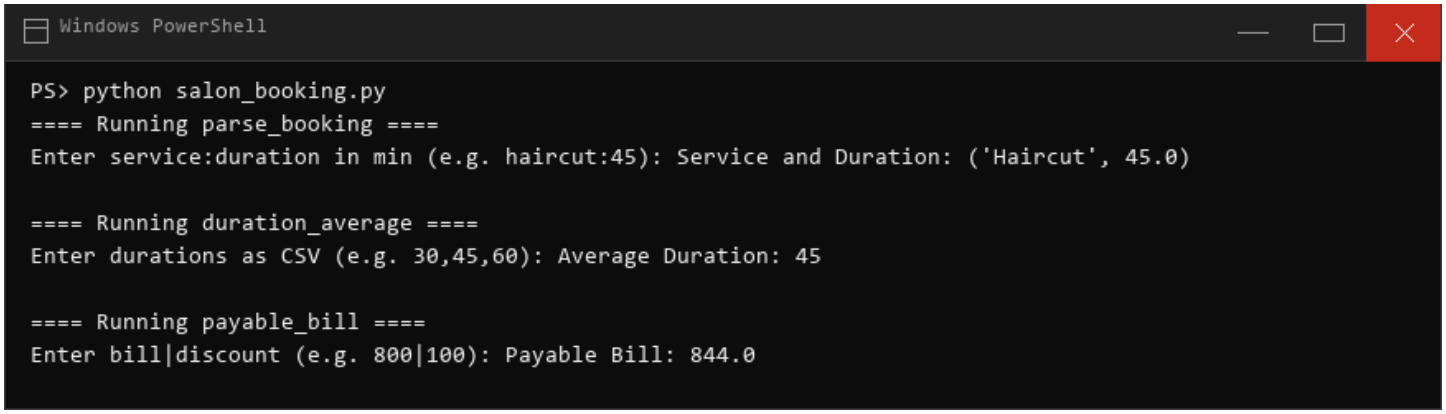
Tool: Hypothesis (property-based fuzz testing for Python)

Repository: github.com/akshat-k30/SE-Lab6-L6_L7_42_salon_booking

Step 2: Run the Buggy Application

Command: python salon_booking.py

The buggy code runs fine with well-formed inputs but has NO input validation. It will crash on None, empty strings, missing delimiters, or non-numeric values.



Step 4: Run Fuzz Tests Against Buggy Code - ALL FAILING

Command: pytest test_salon_booking.py -v
Hypothesis generates random inputs (including None). All 3 tests FAIL with AttributeError because the buggy functions call .split() on None.

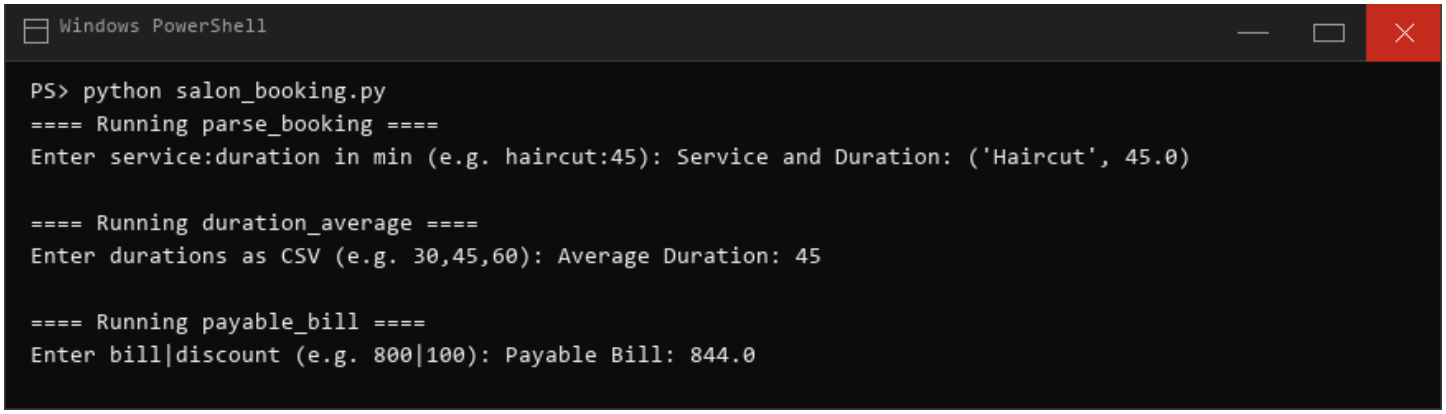
Note: The full pytest output is very long; the screenshot below shows the condensed key lines (header, FAILED markers, error types, and summary).



Step 5: Run the Fixed Application

Command: python salon_booking.py

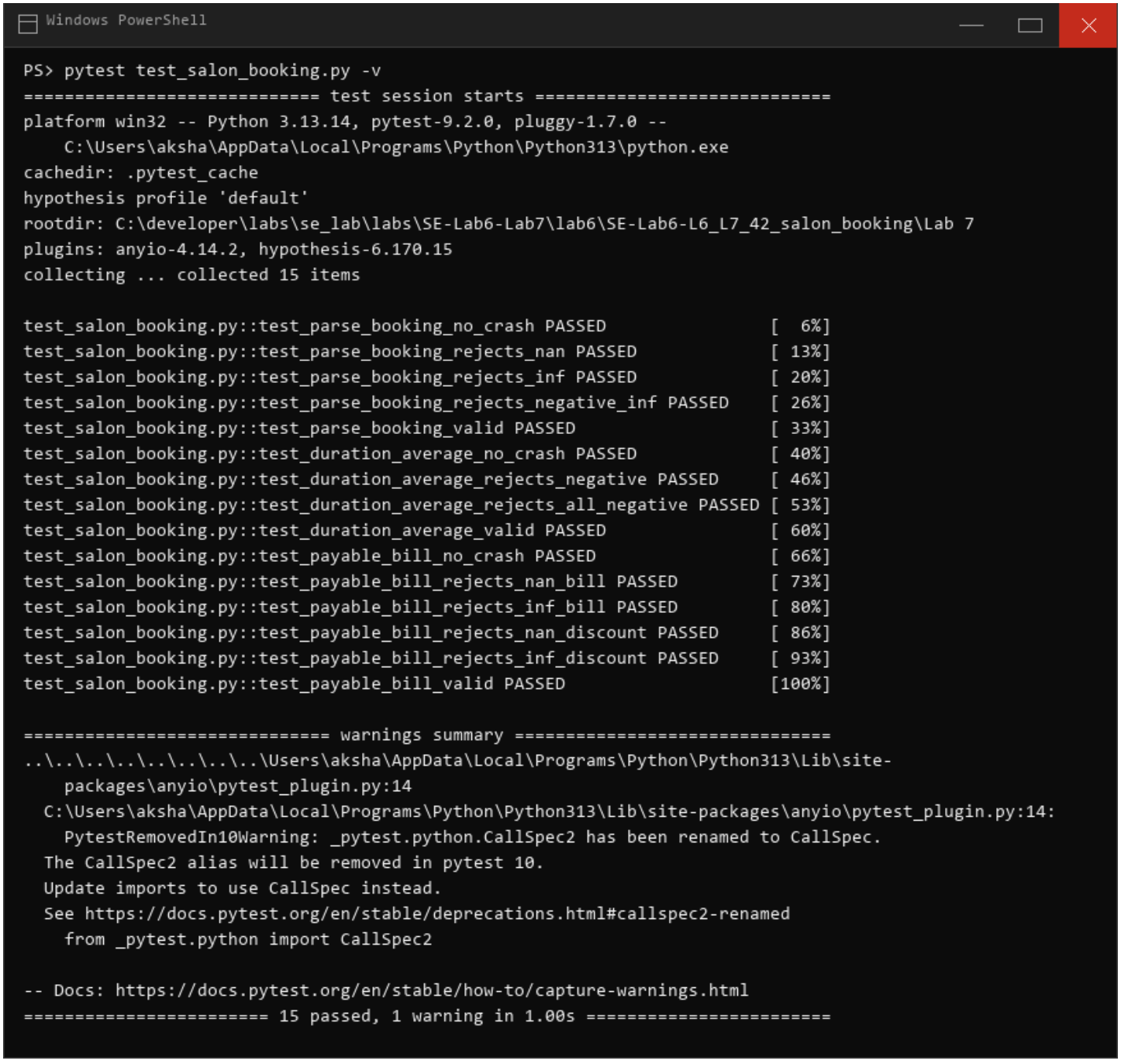
All three functions have been hardened with type checks, delimiter validation, empty-value detection, numeric parsing guards, and range validation. The fixed code produces the same correct output for valid inputs.



Step 6: Re-Run Fuzz Tests Against Fixed Code - ALL PASSING

Command: pytest test_salon_booking.py -v

All 15 Hypothesis fuzz tests now PASS. The test suite includes 3 property-based fuzz tests (one per function) plus 12 explicit edge-case tests covering negative durations, NaN, infinity, and valid-input verification.



Summary of Bugs Found and Fixes Applied

Bugs Discovered via Fuzz Testing:

1. parse_booking(None) --> AttributeError (no null check)
2. parse_booking('haircut') --> ValueError (missing ':' delimiter)
3. parse_booking('haircut:abc') --> ValueError (non-numeric duration)
4. parse_booking(':45') --> returns empty service name
5. parse_booking('haircut:nan') --> accepts NaN as valid duration
6. parse_booking('haircut:inf') --> accepts infinity as valid duration
7. duration_average(None) --> AttributeError (no null check)
8. duration_average('') --> ZeroDivisionError (empty list, div by zero)
9. duration_average('30,,60') --> ValueError (empty CSV element)
10. duration_average('30,abc,60') --> ValueError (non-integer value)
11. duration_average('30,-5,60') --> accepts negative durations
12. payable_bill(None) --> AttributeError (no null check)
13. payable_bill('800') --> ValueError (missing '|' delimiter)
14. payable_bill('abc|100') --> ValueError (non-numeric bill)
15. payable_bill('nan|10') --> accepts NaN as valid bill
16. payable_bill('800|inf') --> accepts infinity as valid discount

Fixes Applied (Defensive Programming):

- Type checking: reject None and non-string inputs with ValueError
- Delimiter validation: check for required separators (':', ',', '|')
- Empty value detection: reject empty fields after splitting
- Numeric parsing guards: try/except around int()/float() conversions
- Finite number checks: reject NaN, inf, -inf using math.isfinite()
- Range validation: reject negative durations, negative bills/discounts
- Graceful error handling in main() with try/except for user input

Test Suite Summary:

15 total tests: 3 Hypothesis property-based fuzz tests (one per function) plus 12 explicit edge-case tests verifying rejection of NaN, infinity, negative values, and correct results for valid inputs. All 15 tests pass - the code handles every edge case gracefully.